AIM

PT.110 PERFORMANCE TEST ENVIRONMENT

<Company Long Name>

<Subject>

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Introduction

Purpose

The purpose of the **Performance Test Environment** document is to describe the details of the technical environment for use in the <Company Short Name> performance test project. It includes:

- definition of the tiering architecture that will be used
- standards used for describing the physical architecture
- strategy for designing the physical database including major operational factors, use of striping and disk arrays (RAID), efficient utilization of disk space and facilitation of database administration tasks
- specification of Key Server Parameters
- mapping of the logical database objects to tablespaces and physical datafiles
- estimation of the I/O throughput requirements
- execution steps required for a single performance test cycle

Environment Hosting

The testing environment is owned, managed and controlled by <Company Long Name>. This data center is located at: <Data Center Location>

Sharing of Environments

The testing activities will share the same environment used by:

- [ ] Business Process Architecture design, Business Requirements Definition
- [ ] Business Requirements Mapping
- [ ] Application and Technical Architecture design/development
- [ ] Module Design and Build
- [ ] Data Conversion development and testing
- [x] Performance Testing
- [ ] Project Team Learning (Training)
- [ ] User Learning (Training)
- [ ] Production

Scope and Application

Performance Test Environment Checklist

The following checklist documents the steps required to prepare the performance test environment.

Seq	Step	Date Completed	Completed By	Comments
1.	Install performance test servers.			
2.	Install operating system on servers.			
3.	Connect servers to network.			
4.	Create user accounts on servers.			
5.	Configure file system on servers.			
6.	Install and connect to network user machines in test environment (if applicable)			
7.	Install Oracle database on servers.			
8.	Install Oracle Applications on servers.			
9.	Migrate database files and data from performance test database development environment.			
10.	Startup database and verify operation.			
11.	Startup Application Concurrent Managers.			
12.	Verify functioning of Oracle Applications and Database functioning.			
13.	Install automated test software (if applicable).			
14.	Verify functioning of automated test software (if applicable).			
15.	Verify execution of transaction programs against test database.			
16.	Develop any custom tools or scripts that you will use for monitoring the tests.			
17.	Install performance monitoring tools.			
18.	Tune test environment.			
19.	Document test cycle execution steps.			
20.	Document baseline Performance Test Environment configuration parameters.			
21.	Configure concurrent manager program execution for programs that execute test transactions.			
22.	Update Site Log for new Applications installation			
23.				
24.				
25.				

Baseline Test Configuration Parameters

In this section we have documented the initial values of the main test configuration parameters prior to performing the test. The test configuration parameters include both parameters that associated with the system, such as database tuning parameters, operating system parameters, etc., but also includes parameter settings for the test itself.

At least some of the these parameters will change during the test for tuning purposes and this is the reason for the designation of the parameters as baseline.

Database Tier

The database tier holds all data and data-intensive programs, and processes all SQL requests for data. The database tier includes the Oracle Database Server, the administration server, and the concurrent processing server. The concurrent processing server executes traditional batch processing tasks such as transactional interfaces and reports.

By definition, platforms in this tier do not communicate directly with applications users, but rather with machines on the application tier that mediate these communications, or with other servers on the database tier.

Database Servers

The database server contains the transactional and schema data associated with the Oracle Applications™. The platform that supports this server contains two sets of objects. One set is the necessary files that support a running database server, and the other set is a series of database files that contain the actual database data.

The following Oracle Database Server instances will be established to support the testing activities of the project:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
PTEST	Compaq Proliant 640 (trn1)	Windows 2000 Server	Oracle 9.5*i*	Performance Testing Environment

Naming Standards

This section describes the naming standards that will be used for objects and files relevant to the database architecture:

- File System Mount Points
- Oracle File Categories
- Database Files

- Tablespaces

File System Mount Points

The standard for File System Mount Points is:

```
Standard:   '/u[01-99]/'
Example:   '/u03/'
```

File Categories

The standard for File Categories is:

```
Standard:   ''
Example:   ''
```

Database Files

The standard for Database File Names is:

```
Standard:   ''
Example:   ''
```

Tablespaces

The standard for Tablespace Names is:

```
Standard:   ''
Example:   ''
```

Mount Points

The standard for File System Mount Points is:

```
Standard:   ''
Example:   ''
```

Oracle Database Instance - <ORACLE_SID>

Key Init.ora Listing

The following table documents the values of other key database init.ora parameters:

Parameter Name	Parameter Value	Description
_undo_optimizer_changes	TRUE	

Additional Database Server Configuration Details

Below are additional configuration details for the Database Server:

Parameter	Value
What is the platform DNS name	ptest.sun02.net
What Type of Hardware Platform and OS is being used	Sun Solaris 9.5
Oracle Block Size	
Net8 Listener Port	1521
Net8 Report Procedure Call (RPC) Server Port	1522
Character Set used for database creation	WE8ISO8859P1

Parameter	Value
Location of Oracle Home Directory	/u01/app/oracle/product/905
Location of Oracle Base Directory	/u01/app/oracle
Is the Optimal Flexible Architecture Standard Being Used	Yes

Tablespace Partitioning

This section describes the basic partitioning of the tablespaces within the production database instances. The partitioning is designed to satisfy <Company Short Name>'s database architecture strategy.

Segment Type	Tablespace	Comments
System		
Temporary		
Rollback		
Tools		
Misc User		
Data		
Index		

Tablespace Storage Parameters

The following table shows the tablespaces for this Oracle Applications database instance:

Tablespace	Size (Mb)	Storage Parameters init	nxt	pct	min	max	Number of Datafiles	Filesystems /u1	/u2	/u3	/u4	/u5
Redo Logs												
SYSTEM												
RBS												
TEMP												
USERS												
Totals	0						0	0	0	0	0	0

Tablespace File Mapping

This section describes the mapping of the tablespace structures onto physical storage files.

Tablespace	Size (MB)	Filesystem	Directory	Datafile
RSEGS	105	sd7h	/ora2/ORACLE/proddb	prod_rseg1.dbf
AOLTAB	105	sd4g	/ora3/ORACLE/proddb	prod_aoltab1.dbf

Tablespace	Size (MB)	Filesystem	Directory	Datafile
ARTAB	210	sd4g	/ora3/ORACLE/proddb	prod_artab1.dbf
GLTAB	262	sd4g	/ora3/ORACLE/proddb	prod_gltab1.dbf
POIND	157	sd4g	/ora3/ORACLE/proddb	prod_poind1.dbf
SHAREIND	52	sd4g	/ora3/ORACLE/proddb	prod_sharedind1.dbf
AOLIND	52	sd5g	/ora4/ORACLE/proddb	prod_aolind1.dbf

Administrative Servers

The administration server is the machine from which you maintain the data in your Applications database. There are three types of operations you will carry out here, each using a different program:

Installing and Upgrading the Database

This process is only performed when installing a new release, or upgrading to a new minor or major release.

Applying Applications Database Updates

Most maintenance patches will consist of new files and scripts that update database objects.

Maintaining the Applications Data

Some features, such as MultiLingual Support and Multiple Reporting Currencies, require regular maintenance to ensure updates are propagated to the additional schemas used by these features

Suggestion: The administration server is the most infrequently used, compared to other servers in the Applications multi-tier environment, and has the smallest computing requirements. You therefore should not need to have more than one administration server for your installation.

The following Administrative Servers will be established to support the performance testing activities of the project:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
PTEST2	Sun E-950 (prod1)	Solaris 5.2	Oracle 9.5*i*	Performance Testing Environment administration

Concurrent Processing Servers

Concurrent Processing Servers are dedicated to running background processes (concurrent programs). The following Concurrent Processing Servers will be established to support performance testing:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
PTEST3	Sun E-950 (prod1)	Solaris 5.2	Oracle 9.5*i*	Performance Testing Environment Concurrent Processing

<Platform Name>

Platform Configuration

Describe in detail the configuration of this platform.

Disk Device Map

The worksheet below documents the disk device map on the hardware platform designated as <Platform Name>.

Device	Filesystem	Available Space
C0S0S0	/u1	4,354 Mb
C0S0S1	/u2	654 Mb
		Mb
		Mb
		Mb
		Mb
		Mb
		Mb
		Mb

I/O Subsystem Configuration

Describe the configuration of the I/O subsystem for this platform.

-
-
-

Application Tier

The application servers form the middle tier between the desktop clients and database servers. They provide load balancing, business logic, and other functionality.

Forms Servers

The forms server is a specific type of application server that hosts the Oracle Forms Server engine. The Oracle Forms Server is an Oracle Developer component that mediates between the desktop client and the Oracle Database Server, displaying client screens and causing changes in the database records based on user actions. Data is cached on the forms server and provided to the client as needed, such as when scrolling through multiple order lines. The forms server exchanges messages with the desktop client across a standard TCP/IP network connection.

Suggestion: Load Balancing Among Forms Servers provides automatic load balancing among multiple application servers. In a load-balancing configuration, a single coordinator called the Metrics Server is on one application server. Metrics Clients located on the other application servers periodically send load information to the Metrics Server so it can determine which has the lightest load. When a client issues a request to download the Forms client applet, the Metrics Server provides the name of the least-loaded host for the applet to connect to.

The following Oracle Web Application (Forms) Server instances will be established to support the performance test:

Platform / DNS	Platform O/S	Server Release	Metric	Forms Port	Forms Server Port	Function
Sun E-950 (was1)	Solaris 5.2	2.5	Server	8000	9000	Applications Server, Load Management Machine (Production)
Sun E-950 (was2)	Solaris 5.2	2.5	Client	8000	9000	Applications Server (Production)
Sun E-950 (was3)	Solaris 5.2	2.5	Client	8000	9000	Applications Server (Production)

Attention: Automatic failover capabilities are inherent in this load-balancing system. If an application server becomes unavailable for any reason, the Metrics Server ceases to route requests to the server until it comes back online. While the application server is offline, requests are routed to one of the other application servers.

Oracle Applications File System

The Oracle Applications File system contains all of the forms and executables that execute of the applications tier of the architecture.

Owner of the File System

The owner of the Oracle Applications File System is: <Userid> (Example: applmgr)

Additional Forms Server Configuration Details

Below are additional configuration details for the Forms Server:

New Cartridge Configuration Settings

Parameter	Value
Cartridge Name:	<Oracle SID>
Object Path:	$ORACLE_HOME/lib/f65webc.so
Entry Point:	form_entry

Virtual Paths

Parameter	Value
/Oracle SID	$ORACLE_HOME/lib
Name of DAD to be used:	<Oracle SID>
Protect PL/SQL Agent:	TRUE
Authorized Ports:	<Web Listener Port>

Cartridge Configuration Screen Values

Parameter	Value
baseHTML	$OA_HTML/afsampled.htm
serverPort	<Forms Port>
userid	APPS/APPS@${ORACLE_SID}
fndnam	APPS

Web Servers

The web server is another type of application server, which runs an HTTP listener. The HTTP listener (also called a web listener) is a component of an HTTP server, such as Microsoft Internet Information Server, or Netscape Enterprise Server. This listener accepts incoming HTTP requests (or URLs) from desktop clients, via the web browser or appletviewer. These requests are either immediately processed by returning an HTML document, or are passed on to the Oracle Web Application Server, which also resides on the same platform.

The following Web Servers will be established to support the performance test:

Server Name	Platform / DNS	Platform O/S	Server Release	UDP Port	Admin Port	Function
PTEST4	Sun E-950 (prod1)	Solaris 5.2	5.4	2649	8888	Performance Testing Environment

Additional Web Server Configuration Details

Below are additional configuration details for the Web Server:

Virtual Directories

File System	Directory Flag	Virtual Path
$APPLTMP	NR	/OA_TEMP/
$APPL_TOP/doc/doc/	NR	/OA_DOC/
$APPL_TOP/html/html/	NR	/OA_HTML/
$APPL_TOP/html/html/bin/	CN	/OA_HTML/bin/
$ORACLE_HOME/forms65/java/	NR	/OA_JAVA/
$ORACLE_HOME/forms65/java/oracle/apps/media/	NR	/OA_MEDIA/

Java Cartridge Path

Virtual Path	Application	Physical Path
/OA_JAVA_SERV	JAVA	$ORACLE_HOME/forms65/java

DAD Settings

Parameter	Value
DAD Name:	<Oracle SID>
Database User:	APPS
Identified by:	APPS
Database User Password:	APPS
ORACLE HOME:	$ORACLE_HOME
Net8 Service:	<Oracle SID>
NLS Language:	American

PL/SQL Cartridge Settings

Parameter	Value
Name of PL/SQL Agent:	<Oracle SID>
Name of DAD to be used:	<Oracle SID>
Protect PL/SQL Agent:	TRUE
Authorized Ports:	<Web Listener Port>

PL/SQL Virtual Path

Virtual Path	Application	Physical Path
/<Oracle SID>/plsql	PLSQL	$ORACLE_HOME/ows/6.0/bin

<Platform Name>

Platform Configuration

Describe in detail the configuration of this platform.

Disk Device Map

The worksheet below documents the disk device map on the hardware platform designated as <Platform Name>.

Device	Filesystem	Available Space
C0S0S0	/u1	4,354 Mb
C0S0S1	/u2	654 Mb
		Mb
		Mb
		Mb
		Mb
		Mb
		Mb
		Mb

I/O Subsystem Configuration

Describe the configuration of the I/O subsystem for this platform.

-
-
-

Desktop Client Tier

The desktop client runs a Java applet using a Java-enabled web browser or appletviewer. The applet sends user requests to the forms server and handles such responses as screen updates, pop-up lists, graphical widgets, and cursor movement. It can display any Oracle Applications screen and supports field-level validation, multiple coordinated windows, and data entry aids like lists of values. A web browser or appletviewer manages the downloading and storage of the Forms client applet on each user's desktop. They also supply the Java Virtual Machine (JVM) that runs the Forms client applet.

Security: Security features ensure that the forms server only accepts connections from "certified" Forms clients bearing the Oracle Applications signature. For additional security, all communication between the Forms client applet and forms server is encrypted using the RSA RC4 40-bit standard form of encryption.

Web Browsers

Below are the web browsers that will be supported during performance testing:

- <list Browsers: Name, OS, Version>
-
-

Web Appliances

Web appliances are devices that are capable of supporting the presentation and execution of a Java applet. These may be single purpose devices or mobile devices. Below are the web appliances that will be supported by the project during testing:

<list web appliances: Name, Manufacturer, Model Number>

Applet Viewers

Applet views are run time environments that run on an operating system and present and execute Java applets. Below are the applet viewers that will be supported by the project during testing:

<list applet viewer: Name, OS, Version>

Network Parameters

The following parameters affect the operation of the network:

Parameter	Value	Comments

Parameter	Value	Comments

Batch Application Parameters

The following table lists the batch programs that should be running during the test execution:

Test Transaction Model	Batch Program Name	# of Conc. Workers	Execution Frequency

Test Cycle Execution

Sequence

1. Copy over test baseline database and control files to test environment

2. Start PT database instance:

   $ su - oracle

   $ sqldba lmode=y

   connect internal

   startup open

3. Start up the SQL*Net listener process:

   $ lsnrctl start

4. Start Oracle Applications concurrent manager:

   $ su - applmgr

   $ ~/local/bin/conc_start

5. Check that the requisite applications batch programs are running and cycling with the correct frequency (see below).

6. Review test tool transaction file and change for test run if necessary

7. Set the test parameters to control precise script execution for the test run.

8. Start system performance test monitoring tools

   <include other steps in here>

9. Monitor test and gather data and test measurements

10. Terminate scripts and batch processing when the appropriate number of (transactions/orders) have elapsed

11. Shutdown concurrent managers

12. Shutdown database.

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date